

AMCAT

Module Syllabi

Sunstone Prime | Placement Preparation Series

9 Modules | *ELQ · WriteX · Excel · Automata · Web Development · SQL · Core Java · Cloud Computing · Node.js*

01 · ELQ — English Language Qualifier

AMCAT Modules 1.1 · 2.1 · 2.4/2.5 | Language + Cognitive Sections

ELQ is the foundational assessment package present across all Sunstone programs. It covers three distinct competency areas — English Comprehension, Logical Reasoning, and Quantitative Ability — each independently scored and mapped to recruiter benchmarks.

MODULE OVERVIEW

English Comprehension	Logical Reasoning	Quantitative Ability
Vocabulary (Synonyms, Antonyms)	Deductive Reasoning	Basic Numbers & Number Theory
Contextual Vocabulary	Coding Deductive Logic	Decimals, Fractions, HCF, LCM
Error Identification	Data Sufficiency	Word Problems — Speed, Profit, %
Sentence Improvement & Construction	Directional Sense	Logarithms
Reading Comprehension	Inductive Reasoning	Permutation & Combination
	Analogy & Pattern Recognition	Probability
	Number Series	Algebra

Component 1 · English Comprehension

AMCAT Module 1.1 — English

Evaluates written English proficiency using internationally standardised resources. Tests the ability to understand written text, interpret spoken-word nuance, and communicate effectively through written documents.

Vocabulary	Grammar	Comprehension
Synonyms	Error Identification	Reading Comprehension
Antonyms	Sentence Improvement	Passage-based inference
Contextual Vocabulary	Sentence Construction	Tone & purpose

Vocabulary

- Synonyms — words with the same or similar meaning across general and domain-specific vocabulary
- Antonyms — words with opposite meanings; tested in context as well as in isolation
- Contextual Vocabulary — selecting the most appropriate word given the sentence or passage context
- Idioms, phrases, and collocations that enrich meaning and professional communication

Grammar

- Error Identification — spotting grammatical mistakes in given sentences (tense, agreement, modifiers)
- Sentence Improvement — choosing the grammatically and stylistically better rewrite from options
- Sentence Construction — arranging jumbled parts into a single coherent sentence
- Subject-verb agreement, pronoun consistency, tense coherence, and correct modifier placement

Reading Comprehension

- Short and medium-length passages followed by comprehension and inference questions
- Main idea identification — distinguishing central argument from supporting details
- Tone, purpose, and author's intent — inferring attitude and register from word choice
- Vocabulary-in-context — determining meaning of unfamiliar words from surrounding text

Prep Note: ELQ English is mandatory across all packages. Vocabulary and Error Identification carry the highest question weight. RC rewards speed — practise timed 3-minute passage reads with a focus on first and last sentences of each paragraph.

Component 2 · Logical Reasoning

AMCAT Module 2.1 — Logical Ability

Assesses the capacity to interpret information objectively, perceive and analyse trends, and evaluate assumptions behind arguments. Strongly correlates with technical trainability, soft-skill trainability, and performance in analyst and knowledge-process roles.

Deductive Reasoning	Inductive Reasoning
Coding Deductive Logic	Analogy Pattern Recognition
Data Sufficiency	Classification Pattern Recognition
Directional Sense	Coding Pattern Recognition
Logical Word Sequence	Number Series Pattern Recognition
Objective Reasoning	
Selection Decision Table	
Puzzles	

Deductive Reasoning

Assesses ability to synthesise information and derive logical conclusions from given premises.

- Coding Deductive Logic — applying a defined coding rule to decode or encode letter/number sequences
- Data Sufficiency — determining whether given data is sufficient to answer a question (Yes/No/Both/Neither)
- Directional Sense — tracking movement across directions to determine final position or distance
- Logical Word Sequence — arranging words in a logical or hierarchical order
- Objective Reasoning — selecting the most logical conclusion from a set of given statements
- Selection Decision Table — applying conditional rules to determine the correct category or outcome
- Puzzles — seating arrangements, blood relations, and ranking-based logic problems

Inductive Reasoning

Assesses ability to learn from patterns, examples, or trends — an indicator of creative problem-solving capacity.

- Analogy Pattern Recognition — identifying the relationship between a pair and applying it to find the missing term
- Classification Pattern Recognition — identifying the odd one out from a group based on a hidden rule
- Coding Pattern Recognition — decoding a pattern in alphanumeric or symbolic coding sequences
- Number Series Pattern Recognition — identifying the rule governing a sequence and finding the missing number

Prep Note: Logical Reasoning is time-pressure heavy — practice under timed conditions. Data Sufficiency questions are tricky: the answer is about whether data is sufficient, not what the answer itself is. For number series, always check differences, ratios, and squares first.

Component 3 · Quantitative Ability

AMCAT Modules 2.4 (Non-Technical) & 2.5 (Technical)

Measures numerical ability and accuracy in mathematical problem-solving. Questions range from purely numeric calculations to arithmetic reasoning, percentage analysis, and quantitative analysis. Two variants exist: Non-Technical (General Maths) and Technical (Engineering Maths).

Basic Numbers (Non-Technical)	Word Problems (Non-Technical)	Technical (Engineering)
Decimals and Fractions	Application of Algebra to real world	Divisibility, HCF, LCM
Factorization, HCF, LCM	Direct and Inverse Proportion	Numbers, Decimal Fractions
Basic Algebra	Speed, Time and Distance	Power & Indices
Odd, Even, Prime, Rational Numbers	Profit and Loss	Profit & Loss
Number System basics	Percentage & Age Relations	Simple & Compound Interest
	Mixtures	Time, Speed and Distance
	Miscellaneous	Inverse
		Logarithms
		Permutation & Combination
		Probability

Quantitative Ability — Non-Technical (Module 2.4)

Designed for non-engineering streams. Tests practical numerical reasoning through real-world word problems and foundational number theory.

Basic Numbers

- Number system — integers, fractions, decimals, negative numbers, positive numbers, rational numbers
- Basic operations — addition, subtraction, multiplication, division on the above number types
- Factorization, HCF (Highest Common Factor), LCM (Lowest Common Multiple)
- Odd, even, prime numbers — properties and identification
- Basic algebra — forming and solving simple linear equations

Word Problems (Applied Mathematics)

- Algebra in real-world contexts — translating word problems into equations and solving
- Direct and inverse proportion — ratio and proportionality problems
- Speed, Time and Distance — relative speed, average speed, trains and boats
- Profit and Loss — cost price, selling price, marked price, discount, percentage profit/loss
- Percentage — percentage increase/decrease, successive percentage change, age relation problems
- Mixtures — alligation method, mixing solutions of different concentrations
- Miscellaneous — work and time, pipes and cisterns, partnership problems

Quantitative Ability — Technical (Module 2.5)

Basic Mathematics

- Divisibility rules, HCF and LCM — for both numbers and algebraic expressions
- Numbers and decimal fractions — simplification, comparison, rounding
- Power and indices — laws of exponents, scientific notation, surds

Applied Mathematics

- Profit and Loss — all standard word problem types
- Simple and Compound Interest — SI vs CI comparison, time-rate-principal problems
- Time, Speed and Distance — trains, boats, relative speed, average speed
- Inverse proportion — indirect variation problems

Engineering Mathematics

- Logarithms — log rules, log equations, change of base, natural log
- Permutation and Combination — nPr , nCr formulas; arrangement vs selection problems
- Probability — classical probability, conditional probability, independent events, mutually exclusive events

Prep Note: For Non-Technical: master percentage and profit/loss — they appear in 60%+ of word problems. For Technical: logarithm rules and P&C formulae are the highest-difficulty questions. Always identify whether the problem needs permutation (order matters) or combination (order doesn't).

02 · WriteX — Writing Assessment

AMCAT Module | Communication Skills Section

Evaluates ability to construct clear, grammatically correct, and contextually appropriate written communication at the sentence, paragraph, and short-essay level. Assesses correctness, coherence, tone, and effective use of language for professional workplace communication.

TOPIC OVERVIEW

Written Communication	Sentence Construction	Short Writing Tasks
Email drafting	Sentence completion	Descriptive paragraphs
Formal vs informal tone	Rearranging jumbled sentences	Opinion/argument construction
Subject-verb agreement	Error correction in passages	Summarisation
Paragraph coherence	Appropriate word choice	Logical sequencing of ideas

Written Communication

- Composing professional emails with appropriate subject, salutation, body, and closing
- Identifying and adjusting tone for formal vs informal contexts
- Subject-verb agreement, pronoun consistency, and tense coherence across paragraphs
- Using connectors and transition words to build cohesive paragraphs

Sentence-Level Skills

- Sentence completion — filling in grammatically and contextually correct words or phrases
- Rearranging jumbled sentences to form a logical paragraph
- Identifying and correcting errors within a given passage
- Selecting the most precise and appropriate word from given options

Short Writing Tasks

- Writing a descriptive or explanatory paragraph on a given topic
- Constructing a coherent opinion or argument with supporting points
- Summarising a given passage in fewer words while retaining key meaning
- Arranging ideas in logical sequence to build a structured short essay

Prep Note: WriteX rewards clarity over complexity. Structure paragraphs as: Topic Sentence → Evidence → Explanation → Link. Common errors: passive-voice overuse, comma splices, and pronoun ambiguity.

03 · Excel — MS Excel

AMCAT Module 4.8.4 | Analytics Section

Tests practical working knowledge — not memorisation of menu locations. Assesses how efficiently a candidate uses Excel to solve real-world data problems, covering functions most commonly used across 80% of professional job roles.

TOPIC OVERVIEW

Excel Basics	Data Analysis & Presentation	Advanced Excel
Getting started	Managing worksheets & workbooks	Macros
Formatting cells	Track changes & comments	Sheet Protection
Entering & selecting data	Pivot tables	WHAT-IF Analysis
Inserting & deleting rows/cols	Analyzing & organizing data	GOALSEEK
Page setup	Charting	
Cut, copy, paste	Mathematical formulae	
Drop-down lists	Text / Logical formulae	

Excel Basics

- Workbook and worksheet navigation — moving between cells, sheets, and workbooks
- Cell formatting — number formats, alignment, borders, fill colours
- Data entry — entering, editing, and selecting cells; drop-down validation lists
- Row/column insertion and deletion; page setup for printing

Data Analysis & Presentation

- Pivot Tables — summarising large data sets by grouping, filtering, and aggregating
- Charting — creating and formatting bar, line, pie, and scatter charts
- Mathematical formulae — SUM, AVERAGE, COUNT, MIN, MAX, ROUND, MOD
- Text formulae — CONCATENATE, LEN, LEFT, RIGHT, MID, TRIM, UPPER, LOWER
- Logical formulae — IF, AND, OR, IFERROR, nested IF statements
- Lookup formulae — VLOOKUP, HLOOKUP, INDEX-MATCH

Advanced Excel

- Macros — recording and running basic macros to automate repetitive tasks
- Sheet and workbook protection — password-protecting cells and ranges
- WHAT-IF Analysis — Scenario Manager, Goal Seek, Data Tables
- GOALSEEK — back-calculating input values to reach a target output

Prep Note: Excel questions are scenario-based — apply the right formula to a real-world situation, not just name it. Practise: VLOOKUP vs INDEX-MATCH, nested IFs, and Pivot Table summarisation.

04 · Automata — Programming Skills Assessment

AMCAT Module 4.9.1 | Automata Section

A live coding assessment — candidates write actual code in an embedded editor, compile it, and pass test cases within the assessment environment. Evaluates code correctness, efficiency (time complexity), and quality against industry-defined best practices. Supports C, C++, Java, and Python.

TOPIC OVERVIEW

Code Correctness	Code Efficiency	Code Quality
Passes basic test cases	Time complexity	Naming conventions
Passes edge test cases	Space complexity	Modular structure
Handles corner cases	Algorithm optimisation	Commenting
Output format accuracy	Avoiding brute-force where possible	Industry best practices

Data Structures

- Arrays and strings
- Linked Lists — singly, doubly, circular
- Stacks and Queues
- Trees and Binary Search Trees
- Hash Maps and Hash Sets
- Graphs — adjacency matrix, adjacency list

Algorithms

- Sorting — Bubble, Selection, Insertion, Merge, Quick Sort
- Searching — Linear Search, Binary Search
- Recursion and backtracking
- Dynamic programming basics
- Greedy algorithms
- Graph traversal — BFS, DFS

Programming Constructs

- Control structures — loops, conditionals, switch-case
- Functions and modular programming
- Dynamic memory allocation
- String manipulation
- OOP concepts for Java/C++ — classes, inheritance, polymorphism, encapsulation

Question Types

- Pattern problems — number, star, and character patterns using nested loops
- Array problems — searching, sorting, rotation, subarray problems
- String problems — palindrome check, anagram, character frequency
- Mathematical problems — prime, GCD, LCM, factorial, Fibonacci
- Logical puzzles — number theory, bit manipulation, recursion-based problems

Prep Note: Write, compile, and test code before submission — multiple attempts allowed. Focus on edge cases. A brute-force solution that passes all cases beats an optimised solution with bugs.

05 · Web Development

AMCAT Module 4.2.16 | IT Skills Section

Assesses ability to design, build, and maintain web applications with both static and dynamic content. Covers the full front-end stack — structure (HTML5), presentation (CSS3), interactivity (JavaScript) — along with fundamental web architecture concepts.

TOPIC OVERVIEW

HTML5 & Structure	CSS3 & Styling	JavaScript	Web Concepts
Semantic HTML	Selectors & specificity	Variables & data types	HTTP/HTTPS protocols
Forms and inputs	Box model	Functions & scope	Client-server model
Multimedia elements	Flexbox & Grid	DOM manipulation	REST basics
Data elements	Responsive design	Events & handling	Cookies & sessions
Application APIs	CSS3 animations	AJAX / Fetch API	Web security basics
Document structure	Pseudo-classes	ES6+ features	Browser rendering

HTML5 Essentials

- Document structure — DOCTYPE, head, body, semantic elements (header, nav, main, footer, section, article)
- Forms — input types, validation attributes, labels, fieldsets
- Multimedia — audio, video, canvas, figure/figcaption
- Data elements — data-* attributes; Application APIs (local storage, geolocation, drag-and-drop)

CSS3 & Presentation

- Selectors — type, class, ID, attribute, pseudo-class, pseudo-element, combinators
- Box model — margin, border, padding, content; box-sizing: border-box
- Layout — display properties, position (relative, absolute, fixed, sticky), flexbox, grid
- Responsive design — media queries, viewport meta tag, mobile-first approach
- CSS3 — transitions, animations, transforms, gradients, shadows, border-radius

JavaScript & Interactivity

- Core — variables (var, let, const), data types, type coercion, operators
- Functions — declarations, expressions, arrow functions, callbacks, closures, IIFE
- DOM manipulation — querySelector, getElementById, innerHTML, createElement, addEventListener
- Async JS — setTimeout, setInterval, Promises, async/await, fetch API
- ES6+ — destructuring, spread/rest, template literals, modules

Web Architecture Concepts

- HTTP methods — GET, POST, PUT, DELETE, PATCH; status codes (200, 301, 404, 500)
- Client-server model — how browsers request, receive, and render pages
- Cookies, sessions, and local storage — differences and use cases
- Web security — XSS, CSRF, HTTPS, input sanitisation

Prep Note: Questions are application-heavy — expect output-prediction (what does this CSS produce?) and code completion. DOM manipulation is highest-weight. Master the difference between display:flex and display:grid.

06 · SQL — Structured Query Language

AMCAT Module 4.2.6 | IT Skills Section

Assesses understanding of database querying and manipulation using SQL and PL/SQL — from basic SELECT statements through advanced multi-table joins, sub-queries, and stored procedure logic.

TOPIC OVERVIEW

Basic SQL	Advanced SQL	PL/SQL
SQL Basics & syntax	JOINS (multi-table queries)	Block structure
Simple queries (SELECT)	Sub-queries	Variables & data types
Filtering & WHERE conditions	Set operators (UNION, INTERSECT)	Control structures
Advanced data filtering	DDL (CREATE, ALTER, DROP)	Cursors
Grouping & aggregation	Views	Stored Procedures
Data integrity	Transaction processing	Functions
DML (INSERT, UPDATE, DELETE)	DCL (GRANT, REVOKE)	Exception Handling

DDL & DML

Data Definition Language (DDL)

- CREATE TABLE — defining columns, data types, constraints
- ALTER TABLE — adding/modifying/dropping columns
- DROP TABLE / TRUNCATE — removing tables or data
- Constraints — PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, CHECK, DEFAULT

Data Manipulation Language (DML)

- SELECT with WHERE, ORDER BY, DISTINCT, column aliases
- WHERE operators — =, !=, >, <, BETWEEN, IN, LIKE, IS NULL
- INSERT INTO, UPDATE with conditions, DELETE with WHERE filtering

Grouping & Aggregation

- GROUP BY, HAVING (filtering after aggregation, not before)
- Aggregate functions — COUNT, SUM, AVG, MIN, MAX

Advanced SQL

JOINS

- INNER JOIN, LEFT/RIGHT OUTER JOIN, FULL OUTER JOIN, SELF JOIN, CROSS JOIN

Sub-queries & Set Operations

- Correlated vs non-correlated sub-queries; sub-queries in SELECT, WHERE, FROM
- UNION (removes duplicates), UNION ALL (keeps duplicates), INTERSECT, MINUS/EXCEPT

Views & Transactions

- CREATE VIEW — virtual table built on a SELECT query
- COMMIT, ROLLBACK, SAVEPOINT — ACID properties
- GRANT and REVOKE — DCL permissions management

PL/SQL

- Block structure — DECLARE, BEGIN, EXCEPTION, END sections
- Variables, constants, %TYPE / %ROWTYPE declarations
- Control structures — IF-THEN-ELSE, CASE, LOOP, WHILE LOOP, FOR LOOP
- Cursors — implicit vs explicit; OPEN, FETCH, CLOSE lifecycle

- Stored Procedures (IN/OUT parameters) and Functions (returning values)
- Exception Handling — WHEN OTHERS, named exceptions, RAISE

Prep Note: SQL questions frequently test JOINS and sub-queries together. Practise GROUP BY with HAVING (not WHERE). PL/SQL focuses on block structure, cursors, and exception handling.

07 · Core Java — Entry Level

AMCAT Module 4.2.3 | IT Skills Section

Evaluates candidates with 1–5 years of Java experience or fresh graduates with strong foundational knowledge. Tests basic constructs through OOP concepts to threading and generics. Questions are output-prediction, error-identification, and application-based.

TOPIC OVERVIEW

Basic Java Constructs	OOPs, Files & Exception Handling	Thread, Generics & Inner Class
Data types & variables	Inheritance	Packages
Classes & objects	Interfaces	Inner classes
Arrays	Files and streams	Strings
Operators	Exception handling	Threads
Control structures	Abstract classes	Generics
Methods	Polymorphism	Collections basics

Basic Java Constructs

- Data types — primitive (int, float, char, boolean, double, long, byte, short) and wrapper classes
- Variables — declaration, initialisation, scope (local, instance, static)
- Operators — arithmetic, relational, logical, bitwise, ternary, instanceof
- Control structures — if-else, switch-case, for, while, do-while, break, continue
- Arrays — single and multi-dimensional; array methods; enhanced for loop
- Classes and Objects — constructors, access modifiers, static vs instance members

Object-Oriented Programming (OOP)

- Inheritance — extends keyword, method overriding, super keyword
- Polymorphism — compile-time (overloading) and run-time (overriding)
- Abstraction — abstract classes, interface implementation
- Encapsulation — private fields, public getters/setters, data hiding
- Interfaces — default and static methods (Java 8+), functional interfaces
- Files and Streams — FileReader, FileWriter, BufferedReader, InputStream/OutputStream

Exception Handling

- Exception hierarchy — Throwable → Error / Exception → Checked / Unchecked
- try-catch-finally blocks; multiple catch blocks; catching by parent type
- throw vs throws; Custom exceptions — extending Exception or RuntimeException
- Common exceptions — NullPointerException, ArrayIndexOutOfBoundsException, ClassCastException

Threads, Generics & Inner Classes

- Strings — String vs StringBuilder vs StringBuffer; common String methods
- Inner Classes — member inner class, static nested, anonymous, local class
- Threads — Thread class vs Runnable; start(), run(), sleep(), join(), synchronisation basics
- Generics — generic classes and methods, bounded type parameters (<T extends Number>)
- Collections — ArrayList, LinkedList, HashMap, HashSet (basic usage and differences)

Prep Note: Focus on output-prediction questions. Key traps: String immutability, == vs .equals(), integer overflow, abstract class vs interface. Threading questions focus on lifecycle states, not advanced concurrency.

08 · Cloud Computing

AMCAT Module 4.3.4 | IT Infrastructure Section

Assesses foundational knowledge of cloud technology — cloud concepts, service models, deployment models, and virtualisation. Designed for awareness-level screening. Questions combine theoretical, conceptual, and applied understanding of cloud infrastructure.

TOPIC OVERVIEW

Cloud Basics	Virtualisation	Private Cloud
Cloud Basic Concepts	Basics of Virtualisation	Basics of Private Clouds
Cloud Service Models	Virtualisation for Enterprise	Private Cloud Platforms
Cloud Deployment Models	Hypervisor types (Type 1 vs 2)	On-premise vs hosted
Key characteristics	VMs vs Containers	Hybrid cloud basics
Cloud economics	Resource pooling	Security in private cloud
Adoption drivers	Snapshots & migration	Private vs public cloud

Cloud Fundamentals

Core Characteristics (NIST Model)

- On-demand self-service — provision resources without human interaction
- Broad network access — accessible via standard mechanisms over the network
- Resource pooling — shared, multi-tenant resource pools
- Rapid elasticity — scale up/down dynamically with demand
- Measured service — usage monitored, controlled, reported (pay-per-use)

Cloud Service Models

- IaaS — virtual machines, storage, networking (e.g., AWS EC2, Azure VMs)
- PaaS — development environments, databases, middleware (e.g., Google App Engine, Heroku)
- SaaS — ready-to-use applications over the web (e.g., Gmail, Salesforce, Office 365)
- FaaS / Serverless — function-based execution, no server management (e.g., AWS Lambda)

Deployment Models

- Public Cloud — third-party owned; shared infrastructure
- Private Cloud — dedicated infrastructure; greater control and security
- Hybrid Cloud — public + private with data/application portability
- Community Cloud — shared among organisations with similar requirements

Virtualisation

- Hypervisors — Type 1 (bare-metal: VMware ESXi, Hyper-V) vs Type 2 (hosted: VirtualBox)
- Virtual Machines — full OS isolation; higher resource overhead; strong security boundaries
- Containers — lightweight, share host OS kernel; faster startup; Docker, Kubernetes
- Snapshots — capturing VM state at a point in time for backup/recovery
- Live migration — moving a running VM between hosts with minimal downtime

Key Cloud Concepts

- Scalability vs Elasticity — planned capacity growth vs automatic on-demand scaling
- High Availability and Disaster Recovery — RTO and RPO concepts
- SLA — guaranteed uptime, performance metrics, vendor obligations
- Cost models — pay-as-you-go, reserved instances, spot instances; CAPEX vs OPEX
- Multi-tenancy — multiple customers share physical infrastructure while data is isolated

Prep Note: Cloud Computing is concept-heavy — no coding. Focus on IaaS/PaaS/SaaS with real examples, the 5 NIST characteristics, and scalability vs elasticity. Virtualisation questions typically compare Type 1 vs Type 2 hypervisors.

09 · Node.js

AMCAT Module 4.2.46 | IT Skills Section

Assesses server-side JavaScript using Node.js. Covers the event-driven, non-blocking I/O architecture, file system operations, asynchronous programming patterns, Express framework, database connectivity, and testing.

TOPIC OVERVIEW

Basics, Events & Streams	File System, Error & Concurrency	DB Handling, Express & Server
Node.js Basics	Buffer and File System	Database Connectivity
Events & Streams	Error Handling	Server creation
Event Loop	Concurrency in Node.js	Express Framework
Non-blocking I/O	Callbacks	Testing in Node.js
npm ecosystem	Promises	REST API basics
Module system	Async/Await	Middleware

Node.js Fundamentals

- Runtime — V8 engine, single-threaded event loop, non-blocking I/O model
- Module system — CommonJS (require/module.exports) and ES Modules (import/export)
- npm — package.json, installing packages, npm scripts, semver versioning
- Core built-in modules — path, os, util, url, querystring, crypto
- Global objects — __dirname, __filename, process, console, Buffer

Events, Streams & File System

- EventEmitter — on(), emit(), once(), removeListener()
- Streams — Readable, Writable, Duplex, Transform; piping streams
- Buffer — creating and manipulating binary data; encoding (utf8, base64, hex)
- File System (fs) — readFile, writeFile, appendFile, readdir, mkdir, unlink
- Synchronous vs Asynchronous file operations — fs.readFile vs fs.readFileSync

Asynchronous Programming & Error Handling

- Callback pattern — callback(err, result) convention; callback hell and mitigation
- Promises — .then() chaining, .catch(), Promise.all(), Promise.race()
- Async/Await — async functions, await keyword, try-catch for async error handling
- Error handling — error-first callbacks, uncaughtException, unhandledRejection
- Concurrency — setImmediate(), process.nextTick(), event loop phases

Express Framework & Server Development

- Routing — app.get/post/put/delete(); route parameters; query strings
- Middleware — app.use(), express.json(), express.static(), custom middleware
- Request/Response — req.body, req.params, req.query; res.send(), res.json(), res.status()
- Error-handling middleware — (err, req, res, next) signature; centralised error management
- REST API design — resource naming, HTTP status codes, JSON response structure

Database Connectivity & Testing

- Connectivity — MySQL (mysql2), MongoDB (mongoose), PostgreSQL (pg)
- Mongoose — schemas, models, CRUD operations, validation, population
- SQL with Node.js — parameterised queries, connection pooling, SQL injection prevention

- Testing — Mocha, Chai, Sinon, Supertest; environment variables with dotenv

Prep Note: Focus on the event loop and async patterns — know callbacks vs Promises vs async/await. Express routing and middleware are high-frequency. Practise writing a basic REST API with CRUD endpoints.